

Math 514, F19
Quiz 1

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (6 points) If two balls are withdrawn from a bag that contains 9 red balls and 6 blue ones.

(a) What is the probability that both balls are red?

$$P\{\text{red, red}\} = \frac{9}{15} \cdot \frac{8}{14} = \frac{12}{35}$$

(b) What is the probability that one is red and the other is blue?

$$P\{\text{red, blue}, (\text{blue, red})\} = \frac{9}{15} \cdot \frac{6}{14} + \frac{6}{15} \cdot \frac{9}{14} = \frac{18}{35}$$

(c) What is the probability of the color being red given that both balls are of the same color?

$$A = \{\text{red, red}\} \quad B = \{\text{red, red}, (\text{blue, blue})\}$$

$$AB = \{\text{red, red}\}$$

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{12/35}{12/35 + \frac{6}{15} \cdot \frac{5}{14}} = \frac{12/35}{12/35 + 1/7} = \frac{12}{17}$$

Problem 2. (4 points) Four buses carry 28, 19, 22 and 31 passengers respectively. One of the total 100 passengers is randomly chosen and let X denote the number of passengers on the bus of the selected passengers.

(a) Find $P\{X > 25\}$.

$$S = \{\text{Bus 1, Bus 2, Bus 3, Bus 4}\}.$$

$$P\{X > 25\} = P\{\text{Bus 1, Bus 4}\} = \frac{28}{100} + \frac{31}{100} = \frac{59}{100}$$

(b) Find $E[X]$.

$$E[X] = \sum_{i=1}^4 x_i P\{X = x_i\}$$

$$= 28 \cdot \frac{28}{100} + 19 \cdot \frac{19}{100} + 22 \cdot \frac{22}{100} + 31 \cdot \frac{31}{100}$$

$$= \frac{2590}{100} = 25.9.$$